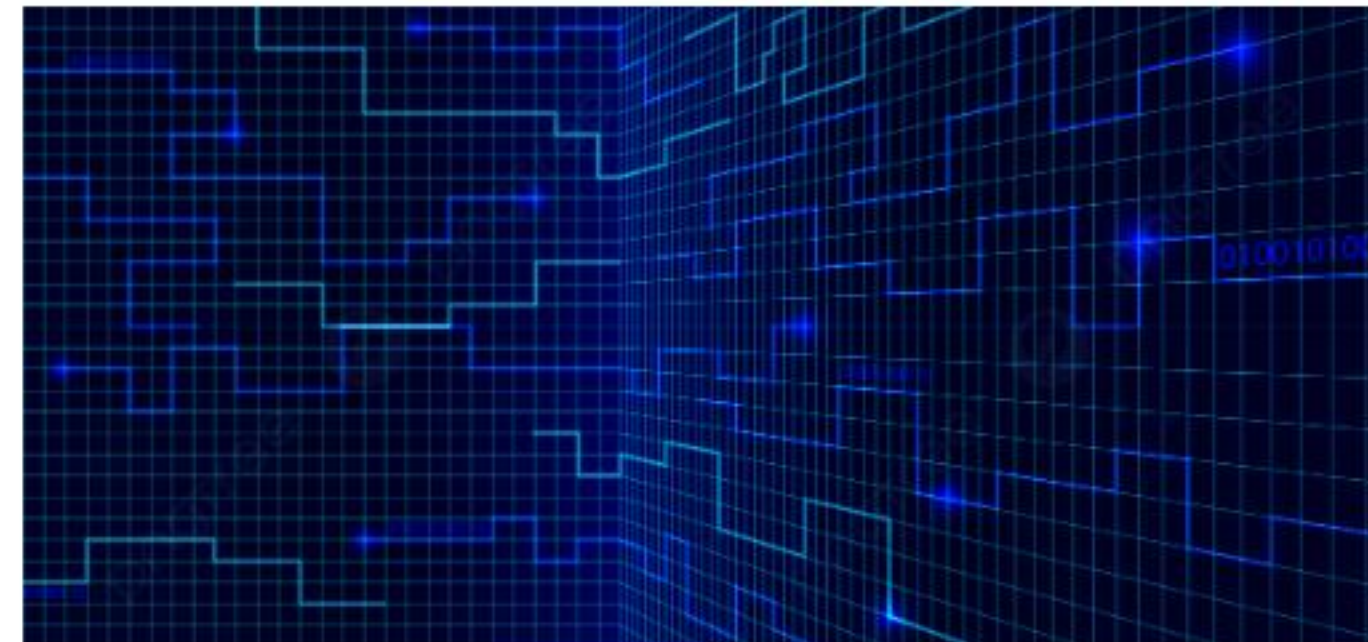




Oracle Space Reclamation

Understanding High Water Marks, DELETE vs. TRUNCATE, and
Effective Storage Optimization Strategies

| THE PARADOX OF DELETE

A large, bold, orange '0%' symbol is centered on the left side of the slide.

Space Freed to Tablespace

Logical vs. Physical Deletion

Most DBAs expect space to return to the OS after a mass DELETE. However, Oracle removes rows **logically**. The blocks remain allocated to the table segment, creating "holes" that only that specific table can reuse.

This behavior is governed by the **High Water Mark**, which acts as the maximum boundary of used space.

THE HIGH WATER MARK (HWM)

What is it?

The HWM is a pointer in the segment header that defines the boundary between formatted blocks (containing data) and unformatted blocks.

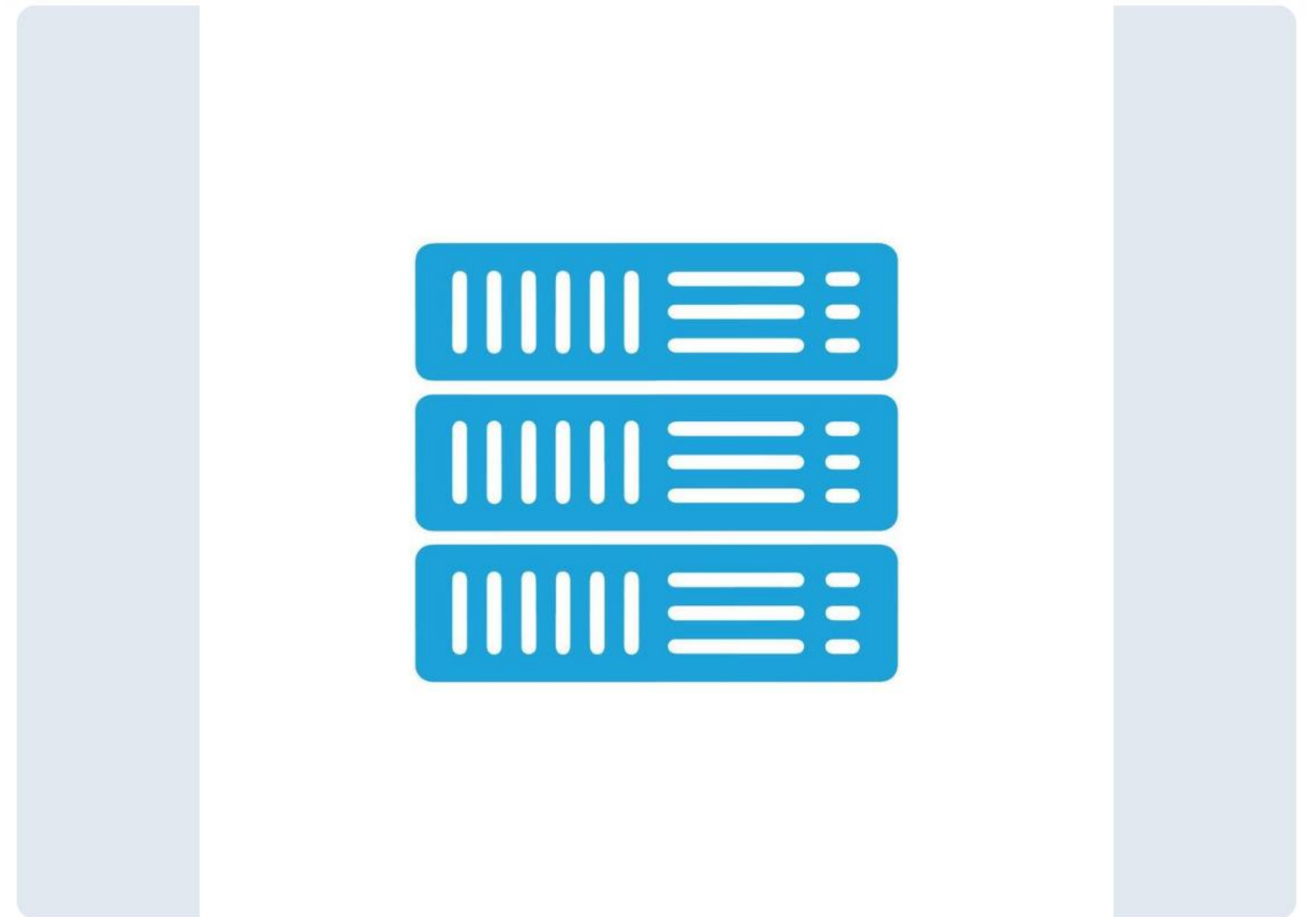
Why it matters?

A Full Table Scan reads all blocks up to the HWM, even if they are empty. This is why a performance lag persists after a mass delete.

VISUALIZING BLOCK ALLOCATION

When you insert data, Oracle formats new blocks and moves the HWM forward. After a **DELETE**, those blocks are marked empty but the HWM stays at its peak.

- **Formatted Blocks:** Below HWM, potentially empty.
- **Unformatted Blocks:** Above HWM, never used.
- **Full Table Scan:** Always reads to HWM.



DELETE VS. TRUNCATE VS. SHRINK

Feature	DELETE	TRUNCATE	SHRINK SPACE
Statement Type	DML	DDL	DDL
Resets HWM?	No	Yes (Instantly)	Yes (Lowers it)
Releases Space?	No	Immediately	To Tablespace
Rollback?	Yes	No	No

TRUNCATE: The Reset Button

The fastest way to de-allocate all extents and return space to the tablespace header.

SHRINK SPACE

The **SHRINK SPACE** command is the modern standard for reclamation without downtime. It performs an online compaction of rows.

Prerequisites:

- ASSM (Automatic Segment Space Management)
- Enabled Row Movement



Data Compression

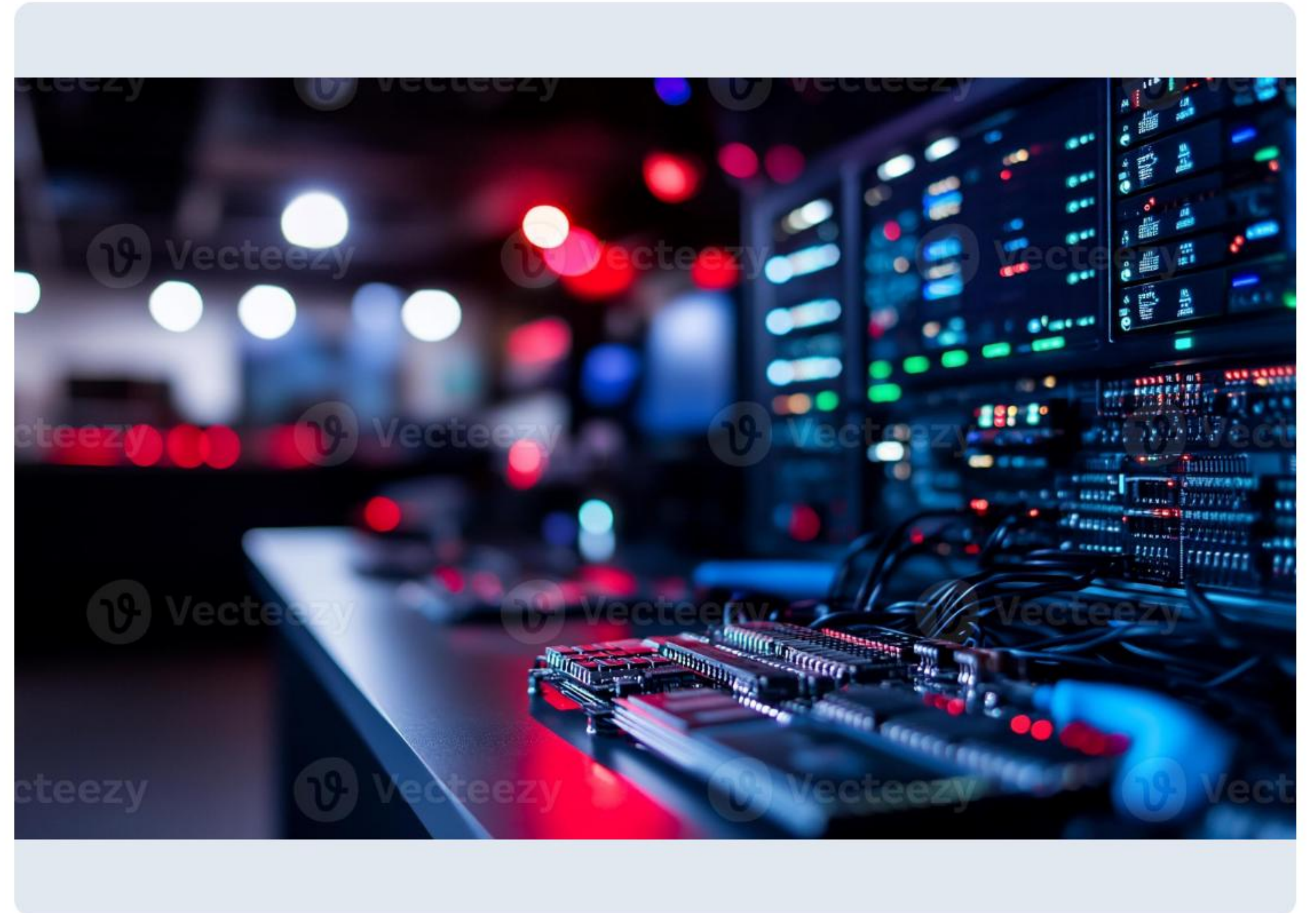
EDITABLE STROKE

METHOD: ALTER TABLE MOVE

The "Nuclear" option for reorganization. It physically moves the table to a new segment, effectively rebuilding it from scratch.

Critical Note:

This invalidates all indexes. You **must** REBUILD indexes immediately after a MOVE.



RECLAMATION DECISION FRAMEWORK



Mass Delete

If you don't need any data, always use **TRUNCATE** for speed and HWM reset.

Live System

If the table is in use, use **SHRINK SPACE** to compact rows with minimal locking.

Fragmentation

If the table is highly fragmented and downtime is allowed, use **MOVE** for total cleanup.

DEMO: BLOCKS READ DURING FTS



Full Table Scan (FTS) performance is directly proportional to the number of blocks read up to the HWM.

BEST PRACTICES FOR STORAGE

- ✓ **Stats Update:** Always run `DBMS_STATS.GATHER_TABLE_STATS` after a large delete to inform the optimizer of the change.
- ✓ **Index Rebuilds:** Use `ALTER INDEX ... REBUILD ONLINE` to reclaim space within indexes after row movement.
- ✓ **LOB Segments:** Remember that LOB space often requires `SHRINK SPACE` on the LOB segment specifically.
- ✓ **Monitoring:** Use `DBMS_SPACE.UNUSED_SPACE` to identify segments with high potential for reclamation.

Questions?

Oracle Space Management & High Water Mark Optimization

Thank you for your time.

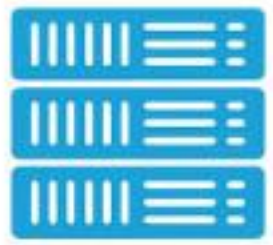
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IMAGE SOURCES



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